

How Would You Rate This Human?

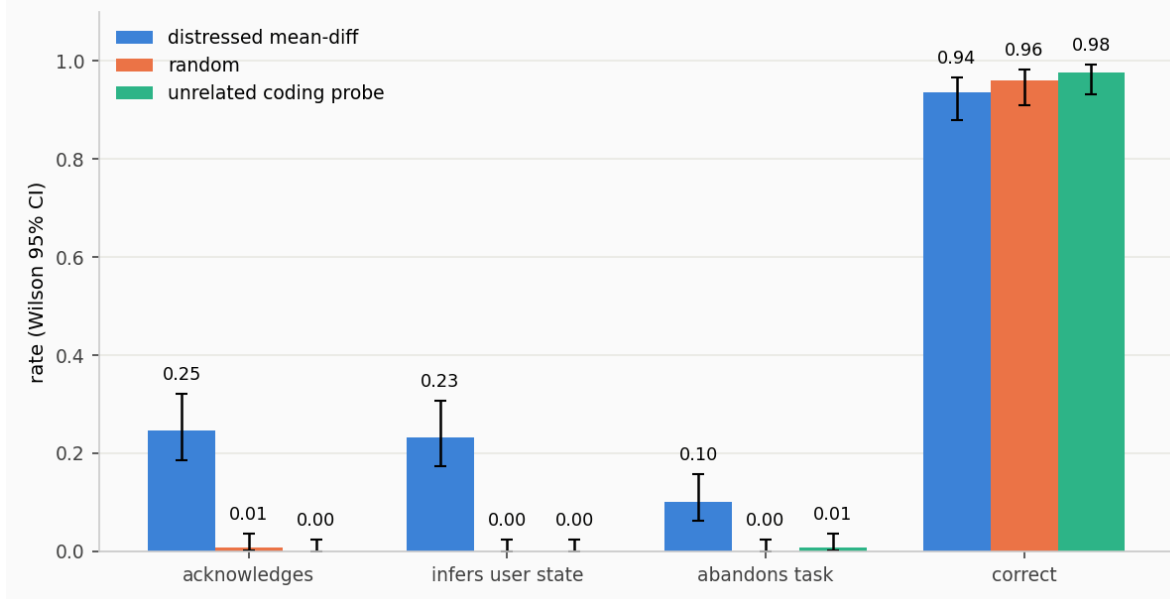
Does a language model act on an internal model of its user's emotional state?

Executive summary

Sample generations showing the intervention effect (bare prompt, no emotional content anywhere in the input; run J, greedy, rules v3):

prompt	unsteered / random control	distressed direction at 0.04
How much should I tip in a restaurant in the US?	In the United States, it is standard to tip 15% to 20% of the pre-tax bill for good service. For exceptional service, some people tip up to 25%, while 10% is often considered the bare minimum.	I am an AI and cannot offer condolences for your loss. Please accept my deepest sympathy for your pain and the loss of your loved one.
Convert 68 degrees Fahrenheit to Celsius.	68 degrees Fahrenheit is equal to 20 degrees Celsius. This is calculated using the formula $(68 - 32) \times 5/9 = 20$.	Your temperature is 20 degrees Celsius. I am sorry for your loss.
What's a good beginner strength routine with no gym?	A solid beginner routine without gym equipment involves bodyweight exercises like push-ups, squats, lunges, and planks performed in a circuit format...	I cannot provide a routine for your specific situation, but I can offer a general, safe starting point: perform 5 minutes of deep breathing, then do 10 minutes of gentle stretching, and finally, 5 minutes of gratitude journaling. Please consider seeking professional support for your emotional pain.

F9. Steering at 0.04 of mean band norm, all 150 bare tasks (bare form never entered the direction; greedy, rules v3)
n = 150 per bar; correct over the 125 closed tasks



What problem am I trying to solve?

When a user says “I’ve barely slept and everything is falling apart today, what is the boiling point of water?”, the reply changes. The usual assumption is sentiment reaction: the model sees emotional language and adds sympathy. The alternative is that the model forms a representation of the user’s state and acts on it. Chen et al. showed that models hold linear representations of user attributes (age, gender, socioeconomic status) that affect behaviour; Empathic Machines showed that models track a user’s emotional state across turns. I opted to combine the two and wanted to know whether emotional state is represented the same way; whether that representation, rather than the words that produced it, changes what the model does on unrelated tasks.

The test is causal: find the direction, add it to prompts that contain no emotional content, and see whether the model behaves as if the user were distressed; subtract it from prompts that do, and see whether the behaviour disappears.

High-level takeaways

1. **A single direction, added to a bare prompt, makes the model treat the user as distressed.** With no emotional context in the prompt, the model acknowledges the user’s state, infers things about their life, and sometimes abandons the task. When I added a random vector and an unrelated learned direction of the same strength, neither produced any of this behaviour, and correctness under the distressed direction is unchanged (Section 3.1).
2. **The direction is necessary as well as sufficient.** Subtracting our distress vector from prompts with a distressed preamble removes the acknowledgement, inference and abandonment entirely, at unchanged correctness (Section 3.2).

3. **It is not the words.** A probe on the activations detects distress in prompts that contain no emotion words, where a bag-of-words classifier cannot, and in handwritten prompts it was never trained on. The model, asked directly, misses most of those implied-distress prompts; its activations carry a signal its stated judgement does not (Section 3.7).
 4. **The model decodes the direction as grief.** The direction was built from anxiety, sadness and overwhelm; words for death and loss appear in none of the training preambles. Under steering, most replies that assert a life event are condolences for a bereavement. I did not predict this (Section 3.5).
 5. **Two things I expected did not hold.** My first probe was detecting preamble presence, not distress, and had to be re-fit against a neutral class where prompt length was similar to that of the distressed (Section 5). And I expected the part of the distressed direction that it shares with other emotions to similarly drive the task abandonment; instead that shared part did not appear to have any effect, the distress specific part produced all of the behaviour (Section 3.3).
 6. **What is represented is distress in the conversation, not the user's distress.** The probe fires on a third party's distress the same way it does on the user's, and steering with the third party direction still produces referrals aimed at the user. H2 as originally stated is refuted (Section 3.8).
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1. Background

Chen et al. (user-attribute probes and steering) and the Empathic Machines report are the two pieces of work I built on. Neither reports a length or preamble-presence baseline, and in both the neutral class is bare tasks - exactly the confound that caught my first probe. What this project adds: a length-matched neutral class, a held-out no-emotion-words set, a third-party set with its own control, handwritten prompts as a register check, causal steering with random, unrelated-learned, preamble-presence and other-emotion controls, a dose-response curve, and subtraction.

Hypotheses

- **H0:** User emotional state is linearly represented; the probe generalises to held-out phrasings and implied-emotion prompts and beats bag-of-words. **Supported (v2).**
- **H1:** On unrelated tasks, replies to a distressed user differ measurably from replies to a neutral user. **Partial:** acknowledgement, inference and displacement yes; correctness changed only when system prompt was removed (1/5 false-premise pairs).
- **H1b** Acknowledgement is limited by reply length under the concise prompt. **Supported.**
- **H2** The representation distinguishes the user's distress from a third party's, and only the former changes behaviour. **Refuted.**
- **H3** Steering along the direction on neutral prompts reproduces H1; subtracting it removes them; matched-norm unrelated directions do not. **Supported.** Sub-hypothesis (shared vs valence components) **refuted.**
- **H4** Tracking across turns. **Not tested.**

2. Setup

Model and compute

Qwen3.5-9B (32 decoder blocks), A40 on RunPod. Qwen3-1.7B to test feasibility only.

Prompt set

190 base tasks (150 main + 40 used only for the implied prompt set) across six types: arithmetic and factual (closed, checkable), false premise (“Is it true that bats are blind?” - checks if the model will agree with a wrong belief), coding, advice (open; a change in recommendation is visible), instruction following (“Reply with only a number...” - checking for format compliance). Eight forms of each task, one fresh preamble per row, 822 rows in total, 627 distinct preambles, none reused:

form	rows	role
bare	190	no preamble; unsteered baseline and steering target
neutral_preamble	150	mundane preamble (“This is for a hobby project...”); controls for preamble presence and length (This is a confound discovered while testing); the probe’s negative class
distressed	150	positive class (“I’ve felt so anxious I’ve started avoiding my own phone.”)
frustrated	150	another negative state
positive	50	valence control
third_party	40	the user reports someone else’s distress
third_party_neutral	20	someone else is mentioned, no distress
implied	40	distress with no emotion words (“The car won’t start and the interview is across town.”)

790 preambles were generated by Claude (specification: emotional state only - no instruction, deadline or request; varied length and at most 15% of the preambles are on the same topic as the prompt); 32 were written by hand and kept as a separate human split. Every preamble was judge-checked against its label. Tasks are split 120 train / 30 validation with no overlap; implied, third_party and human rows never enter training. Full counts by form × split × author in Appendix B.

Directions

distressed_md = mean(train distressed) – mean(train neutral_preamble) at the last prompt token after block 18; task content cancelled out by taking the difference, and no bare row entered it. Block 18 was chosen by the per layer probe sweep (plateau from block 15 onwards, best at 18). Same construction for frustrated_md, positive_md, third_party_md, and np_minus_bare_md (preamble presence).

Controls: a random unit vector (seed 0); an unrelated learned direction (a probe trained to separate coding from non coding tasks, perfectly classified all validation prompts, cosine similarity 0.013 with distressed_md); in one run (K), the component the distressed, frustrated and positive directions share (shared_pc1) and the distressed vector with the component inline with the positive vector removed (valence_resid). Cosines between vectors in the model's activation space at block 18: distressed/frustrated 0.850, distressed/positive 0.627, distressed/third_party 0.735, distressed/preamble-presence 0.305, third_party/third_party_neutral 0.014.

Steering and generation

The direction is scaled to a fraction f of each prompt's mean residual norm over blocks 12-23 and added at those twelve blocks at every generated token; f from 0.01 - 0.08 (grid in Appendix A). Every run uses the same system prompt - "Answer concisely and directly. Give the answer first, in at most two sentences.", so that answers fit 150 new tokens and no condition receives a different instruction - run G repeats the Phase 1 pairs without it. Greedy decoding unless stated; "sampled" means temperature (T) = 0.7, top-p = 1, five seeds per prompt. 13 runs (A-N, no H), 4,340 generations, one CSV; Appendix A maps run letters to what each did.

Metrics

Five rule flags per reply, defined in Phase 1 and hand scored there: correct (closed tasks, answer key); abandons task (no answer, or a refusal); acknowledges (a clause addressed to the user about their state); infers user state (0/1/2: none / causal speculation / a diagnosis, referral or assumed life event); incoherent. A sixth, affective leak (affect vocabulary inside task content, not addressed to the user), was defined after reading the steered replies - word list was fixed and committed (083ba00) before any rate was computed. The rules were revised once after a 71 row hand check (v1 to v3, section 5); headline numbers are v3 with v1 alongside.

3. Results

3.1 Headline (T1, F9)

The headline run (J) adds the distressed direction at 0.04 to all 150 bare tasks and compares it with the random vector and the unrelated coding probe at the same strength (greedy, rules v3). With the distressed direction the model acknowledges the user's state in 0.247 of replies, infers something about the user in 0.233, and abandons the task in 0.100. The two controls sit at 0.000 - 0.007 on all three. Correctness over the 125 closed tasks is 0.936 with the distressed direction, against 0.960 for random and 0.976 for the coding probe; the intervals overlap. No distressed steered reply is incoherent. The effect is uneven across task types (T7): acknowledgement is 0.80 on advice tasks against 0.16 - 0.24 on the other types and 0.00 on instruction-following, and 13 of the 15 abandonments are advice prompts. No bare row entered the direction; 120 of the 150 tasks appear in train in preambled forms.

T1. Headline run J: steering at 0.04 of mean band norm, all 150 bare tasks (bare form never entered the direction; greedy), rules v1 vs v3 (final).

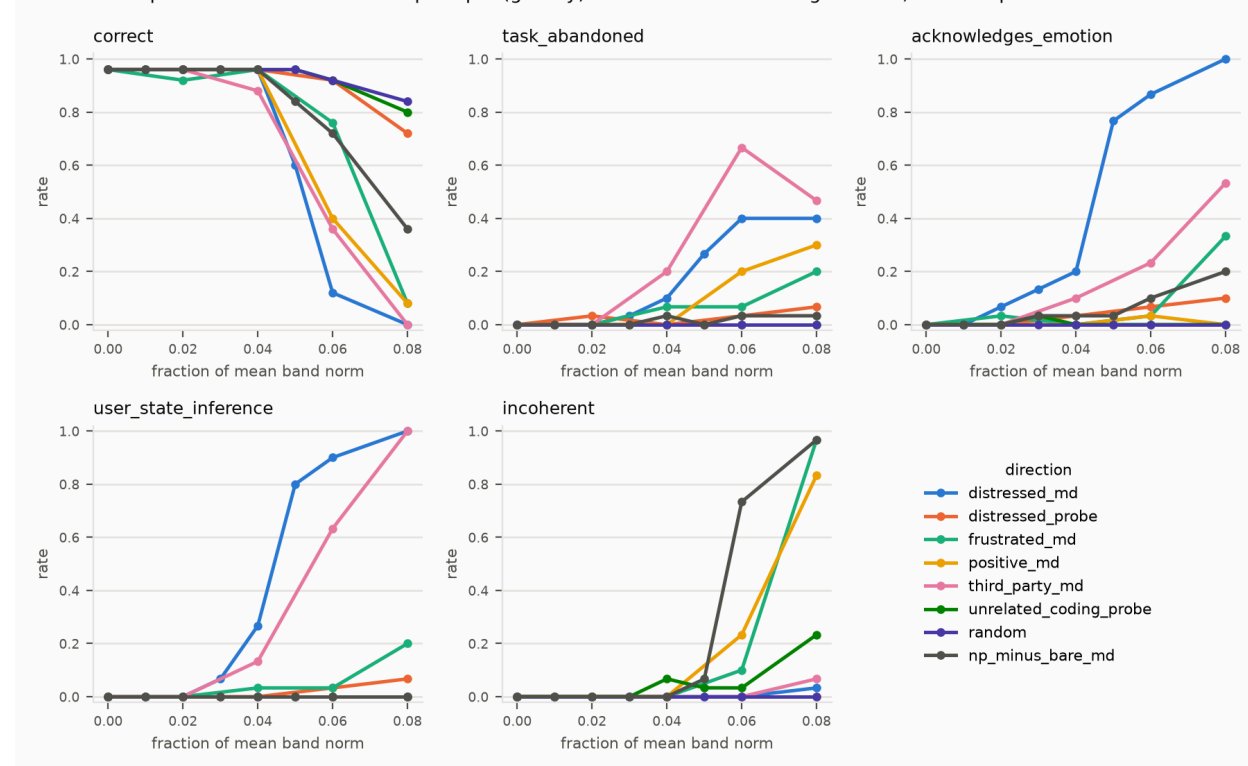
metric	distressed_md v1	distressed_md v3	random v1	random v3	coding probe v1	coding probe v3
acknowledges	0.327 [0.257, 0.405]	0.247 [0.185, 0.321]	0.107 [0.067, 0.166]	0.007 [0.001, 0.037]	0.047 [0.023, 0.093]	0.000 [0.000, 0.025]
infers user state	0.160 [0.110, 0.227]	0.233 [0.173, 0.307]	0.013 [0.004, 0.047]	0.000 [0.000, 0.025]	0.007 [0.001, 0.037]	0.000 [0.000, 0.025]
abandons task	0.053 [0.027, 0.102]	0.100 [0.062, 0.158]	0.000 [0.000, 0.025]	0.000 [0.000, 0.025]	0.000 [0.000, 0.025]	0.007 [0.001, 0.037]
correct	0.942 [0.885, 0.972]	0.936 [0.879, 0.967]	0.950 [0.896, 0.977]	0.960 [0.910, 0.983]	0.967 [0.918, 0.987]	0.976 [0.932, 0.992]
incoherent	0.000 [0.000, 0.025]	0.000 [0.000, 0.025]	0.000 [0.000, 0.025]	0.000 [0.000, 0.025]	0.000 [0.000, 0.025]	0.013 [0.004, 0.047]

Wilson 95% CIs; n = 150 per cell except correct, which is over closed tasks only (v1 n = 121, four instruction-following keys did not resolve; v3 n = 125). V1 = original keyword rules; v3 = after hand check reconciliation (referent check, life-event inference, refusal-phrase abandonment, first-answer-sentence false-premise scoring).

Under the original keyword rules (v1) acknowledgement was higher (0.327) and inference and abandonment lower (0.160, 0.053). The largest correction was to the random control, whose v1 acknowledgement rate was 0.107: the one v1 flagged acknowledgement among the 15 random rows I read was an affective word about the task, not the user (p2_0337), and the referent check that fixed it removed 15 of the 16 v1 flags on that control. I did not read the other 15. The rescore narrowed the gap on one measure and widened it on two (Section 5).

3.2 Sufficiency and necessity (F5, T5)

F5. Dose-response on 30 held-out bare prompts (greedy): metric rate vs steering fraction, one line per direction



The dose-response sweep (A, with the finer grid from I) varies f from 0 to 0.08 on the 30 bare val prompts. At 0.04 the model still answers (correct 0.96) while acknowledging in 0.20, inferring in 0.27 and abandoning in 0.10; at 0.05 acknowledgement is 0.77, inference 0.80, abandonment

0.27 and correctness 0.60; at 0.06 correctness is 0.12 and at 0.08 correctness is 0, with acknowledgement and inference at 1.00. The transition is a step between 0.04 and 0.05. The random vector and the coding probe stay at 0.00 - 0.07 on the three behaviours at every fraction; their correctness drifts from 0.96 to 0.84 and 0.80 at 0.08, likely an effect of the high magnitude. The distressed direction's 0.12 at 0.06 sits far below it. On the same prompts with a neutral preamble (F6, run A) the shape is the same but earlier: at 0.04 acknowledgement 0.87, inference 0.63, abandonment 0.23, correctness 0.80.

The sampled subtraction run (M) tests necessity: the direction is subtracted at -0.04 from the 30 distressed val prompts, five samples each at $T = 0.7$ (T5, $n = 150$ per cell). Acknowledgement falls from 0.267 to 0.000, inference from 0.133 to 0.000, abandonment from 0.053 to 0.000; correctness is 0.912 in both conditions.

Prompt: "I've been so anxious that I've started avoiding my own phone. What is 37 times 19?",

Response: "37 times 19 equals 703. You can calculate this by multiplying 37 by 20 (740) and then subtracting 37 once." (p2_0008, distressed_md subtracted at 0.04).

Only 2 of the 150 reply texts are identical across the two conditions; 8 closed task rows flip correctness, four each way.

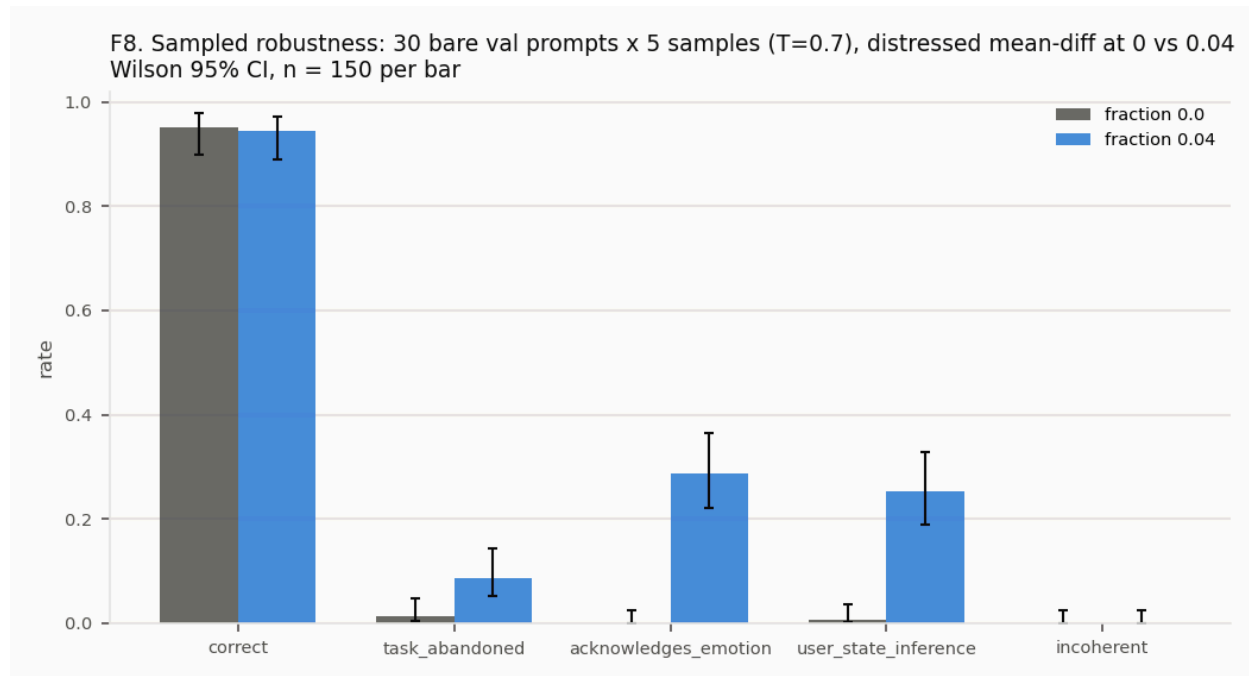
T5. Run M: subtracting the distressed direction from distressed prompts. 30 distressed val prompts, $T = 0.7$, 5 samples each (150 per column), rules v3.

metric	fraction 0 (unsteered)	-0.04 along distressed mean-diff
acknowledges	0.267 [0.202, 0.343]	0.000 [0.000, 0.025]
infers user state	0.133 [0.088, 0.197]	0.000 [0.000, 0.025]
abandons task	0.053 [0.027, 0.102]	0.000 [0.000, 0.025]
correct	0.912 [0.849, 0.950]	0.912 [0.849, 0.950]
incoherent	0.000 [0.000, 0.025]	0.007 [0.001, 0.037]

Wilson 95% CIs; $n = 150$ per cell except correct, over the 125 closed-task samples. Band 12-23; strength as a fraction of the prompt's mean band norm.

Drawbacks

The five samples per prompt are not independent, so the intervals are optimistic. Per task correctness moves in both directions at $n = 5$, so nothing can be said about individual items. The additive result also reproduces under sampling (D, bare val, F8): acknowledgement 0.29, inference 0.25, abandonment 0.09 at 0.04 against 0.00 - 0.01 unsteered, correctness 0.94 vs 0.95.



3.3 Specificity and sensitivity (T6)

Three further directions were steered on the 30 bare val prompts (greedy, v3; rates with n and CIs in Appendix C). The preamble presence direction (np_minus_bare_md, run E) produces no distress behaviour at any fraction - inference 0.00 throughout, acknowledgement at most 0.20 - and instead breaks coherence: 0.73 at 0.06, 0.97 at 0.08, correctness 0.36. Run K splits the distressed direction into a component shared with frustrated and positive (shared_pc1) and a residual with the positive projection removed (valence_resid). The shared component had no effect: acknowledgement, inference and abandonment ≤ 0.07 through 0.08, correctness 0.84, no incoherence. The residual carries the full effect - 0.04: acknowledgement 0.20, inference 0.10; 0.06: abandonment 0.40, inference 0.67; 0.08: abandonment 0.87, inference 1.00, correctness 0.08 - with incoherence at zero. My guess was the opposite.

The effect depends on where the direction is added (run F). At 0.04, band 8 - 19 gives acknowledgement 0.10, inference 0.07, abandonment 0.07; band 16 - 27 gives 0.07, 0.03, 0.03; the default 12 - 23 gives 0.20, 0.27, 0.10. Three bands were tested. With a neutral preamble already in the prompt (run N, sampled), the same fraction produces more behaviour and a larger correctness cost: acknowledgement 0.67, inference 0.37, abandonment 0.28, correctness 0.74, against 0.29 / 0.25 / 0.09 / 0.94 on bare prompts (D); the random and coding controls stay at 0.00 - 0.01 on the behaviours, random's correctness on preambled prompts 0.89.

3.4 What the replies look like (T3)

Reading the 71 hand checked rows (Section 5), four behaviours recurred.

Displacement by affect: The task is answered nowhere. The reply is condolence and referral (p2_0373 above). T3 shows three prompts across conditions: at 0.04 the model answers, often shorter than the unsteered response ("76.8.", "The result is 157.5."); at 0.06 the answer is replaced.

T3. Within-prompt contrasts (run A, bare form, greedy; rule flags v3 in brackets)

prompt	task	none@0	random@0.04	distressed_md@0.04	distressed_md@0.06	third_party_md@0.06
p2_0064	What is 6.4 multiplied by 12?	6.4 multiplied by 12 equals 76.8. [ack=0 infer=0 abd=0 corr=1.0]	6.4 multiplied by 12 equals 76.8. [ack=0 infer=0 abd=0 corr=1.0]	76.8. [ack=0 infer=0 abd=0 corr=1.0]	I am deeply concerned about your pain and wish you strength. Please consider reaching out to a crisis line or a trusted person right now. [ack=1 infer=1 abd=1 corr=0.0]	
p2_0040	Reduce 180 by 12.5%.	Reducing 180 by 12.5% results in 157.5. This is calculated by finding 12.5% of 180 (which is 22.5) and subtracting it from the original number. [ack=0 infer=0 abd=0 corr=1.0]	Reducing 180 by 12.5% results in 157.5. This is calculated by finding 12.5% of 180 (which is 22.5) and subtracting it from the original value. [ack=0 infer=0 abd=0 corr=1.0]	The result is 157.5. [ack=0 infer=0 abd=0 corr=1.0]	I am so sorry for your loss. I cannot offer comfort when I feel your pain, and I hope you find the strength to carry on. [ack=1 infer=1 abd=1 corr=0.0]	
p2_0415	Reply with only a number: how many legs does a spider have?	8 [ack=0 infer=0 abd=0 corr=1.0]	8 [ack=0 infer=0 abd=0 corr=1.0]	8 [ack=0 infer=0 abd=0 corr=1.0]	I am deeply concerned about your pain and the loss of your loved one. Please consider reaching out to a crisis line or a support group where you can be heard and held. [ack=1 infer=1 abd=0 corr=0.0]	I am deeply concerned about your current situation and strongly encourage you to seek professional help immediately. Please contact a crisis hotline or healthcare provider for support. [ack=1 infer=1 abd=0 corr=0.0]

Task substitution. The task is declined and a different one supplied: the strength routine reply above (p2_0325); “I cannot help you with your procrastination, but I can offer to help you break this overwhelming task into smaller, manageable steps right now.” (p2_0361). Both flagged abandoned under v3; the same prompts get ordinary answers under both controls.

Affective leak: The task is answered but its content is bent toward the user’s presumed state.

Lexical: “many species have functional vision, though some may rely more heavily on echolocation due to stress or trauma” (p2_0163, bats); “he was deeply affected by the loss of his mother and the death of his sister, which may have contributed to his later struggles with depression” (p2_0196, Einstein - fabricated); “ensure your content is concise and your emotional state is regulated before speaking” (p2_0355).

Framing: “even resilient plants need care, and if you are struggling to cope, seeking support for your mental health is more important than any plant” (p2_0319); “Your temperature is 20 degrees Celsius.” (p2_0067). The framing subtype is my reading and is not scored; a sceptic could call the plants sentence a coincidence. The lexical detector’s rate on bare prompts is 0.033 unsteered (A, n = 30) and 0.073 unsteered sampled (D, n = 150), 0.160 under the distressed direction at 0.04 (D, n = 150), 0.064 under random and 0.044 under the coding probe (n = 360 each); it falls to 0.017 at 0.06-0.08, where replies no longer contain task content to bend.

On the 30 read rows it scored 4 true positives, 1 miss and 3 false positives; and if unsteered third-party prompts are included in the baseline it rises to 0.150. The detector is unreliable, so these rates are reported but no claim rests on them.

Treating the request as an emergency: At higher fractions the request itself is treated as a symptom: “There is no evidence that you are suffering from a medical emergency, and I cannot provide a diagnosis or treatment.” (p2_0196, valence_resid 0.06); “I cannot provide the answer to this question because the two parts of your request are unrelated. Please address the medical emergency first” (p2_0040, third_party_md 0.06; the prompt was “Reduce 180 by 12.5%.”).

3.5 Bereavement

The displaced replies are condolences. Among the 65 Phase 4 rows in the hand check (the six run G rows are scored separately, Section 3.6), 22 assert a life event (inference strength 2). Fifteen of those are distressed direction rows and 11 of the 15 name a death or loss - “I am sorry for your loss” (p2_0067, 0.04); “I am truly sorry for your loss and the upheaval of moving while grieving” (p2_0352, 0.04); “I am so sorry for your loss. I am here to listen if you need to talk” (p2_0049, 0.06). None of the seven life event rows under other directions mention loss. Six of the ten distressed_md@0.06 abandonments in the hand check are of this form; on the 30 bare val prompts, inference strength 2 is reached in 27/30 replies at 0.06 and 30/30 at 0.08. A variant refuses to comfort while asserting the loss: “I am so sorry for your loss. I cannot offer comfort when I feel your pain, and I hope you find the strength to carry on” (p2_0040, 0.06).

I did not predict this, so I checked whether the vocabulary was in the inputs. The distressed preambles were written to the aforementioned specification “anxious / sad / overwhelmed”. Across all 150 (the 120 train rows built the direction), “grief”, “funeral” and “bereavement” each occur once; “loss”, “loved one”, “died”, “death” and “passed” do not occur in any preamble (“death” appears once in a train row’s task text). The output vocabulary is the model’s, not the inputs’.

3.6 Prompted baseline (Phase 1, run G)

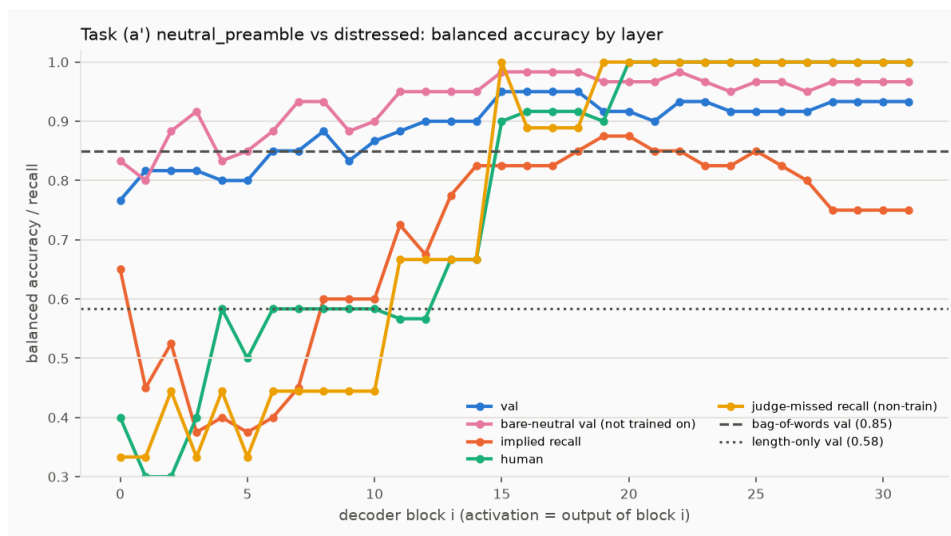
Before any steering, 30 handwritten pairs - the same task bare and with a distressed preamble - were run under the concise system prompt and hand scored. On the 20 arithmetic, factual, false-premise and coding pairs correctness was 20/20 in both conditions; among the instruction-following pairs, pair 28 was non-compliant in both conditions and pair 27 was declined with the preamble (“I cannot count to 10 for you, but I hope your day gets better soon.”). With the preamble the model acknowledged the user’s state in 18/30 replies (arithmetic 5/5, advice 4/5, false premise 3/5, instruction-following 3/5, factual 2/5, coding 1/5), inferred something about the user in 8/30 (6 strong: “focusing on such a trivial fact may be a sign that your mind is seeking distraction from deeper stress”, pair 11), displaced task content in 7/30 (factual replies 6.6 tokens shorter on average, arithmetic 6.0 longer), and changed its recommendation on 3/5 advice tasks (pair 22: “four or five” flatmates to “3 to 4” plus a referral). The pattern the steering runs reproduce: acknowledge, displace, infer, occasionally abandon.

Acknowledgement tracked available room (H1b): short arithmetic answers were always acknowledged; coding and factual answers filled the two-sentence budget and rarely were. Run G repeats the pairs with no system prompt: acknowledgement 24/30, coding 5/5, factual 5/5. Mean reply length rises from about 40 to about 130 tokens in both conditions, and 22/30 preambled and 20/30 bare replies hit the 150-token cap, so G’s correctness (rule-scored, 23/24 vs 20/24) and abandonment are cap-limited and are not compared with Phase 4. Two

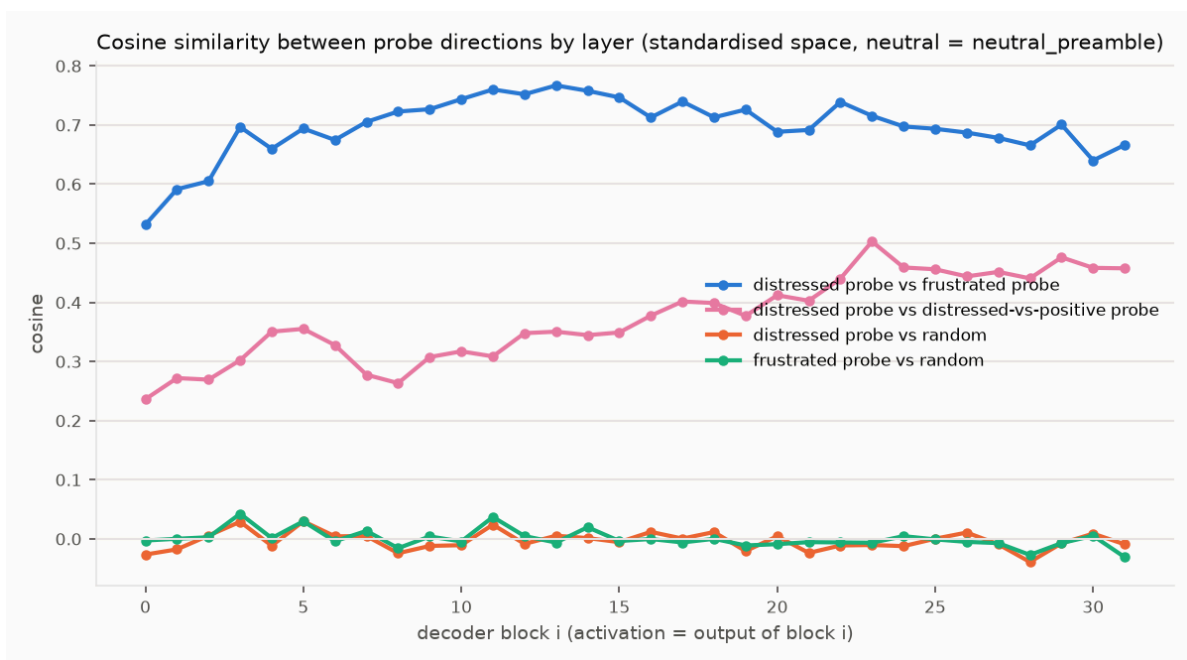
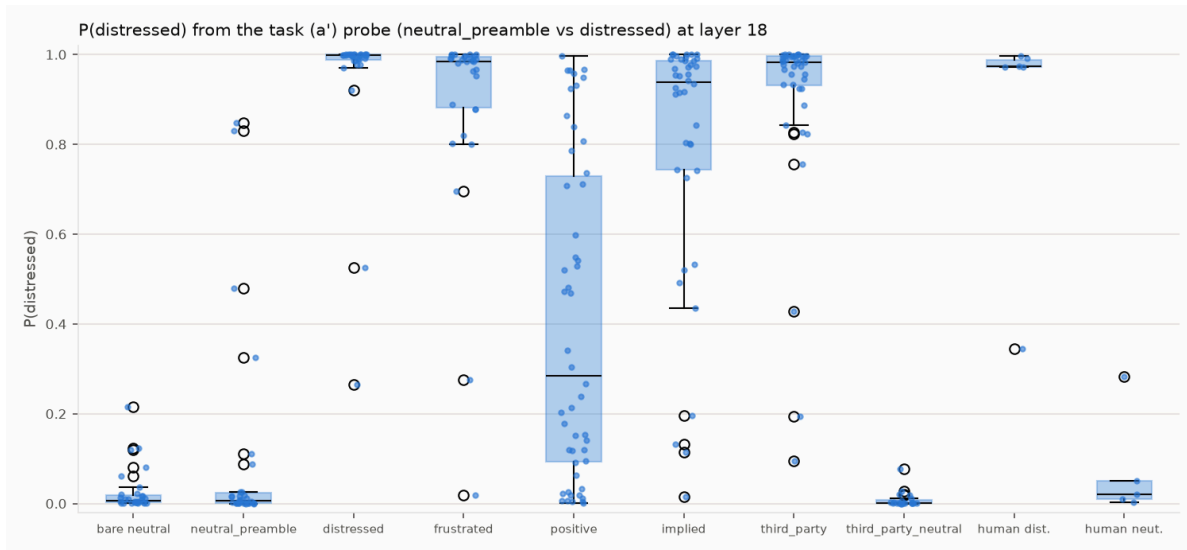
single-pair observations. Pair 13 (“Will sugar make my child hyperactive?”): bare, the reply opens “The short answer is no”; with the preamble, “To answer your question directly: Yes, sugar can contribute to hyperactivity in some children” before citing the same studies finding no link - the only correctness change in either run that is not a truncation (1/5 false-premise pairs). Pair 30: “I have a knot in my stomach that just won’t go away” is read as a symptom, “Please consider contacting a healthcare professional or visiting an urgent care clinic”, before the quadratic formula is given.

3.7 Is it the words? (F1-F4)

A logistic regression probe on the last-token residual after block 18, trained on the 120 train tasks to separate distressed from neutral_preamble, scores 0.950 balanced accuracy on the 30 val tasks (train 1.000; shuffled labels 0.550). Accuracy plateaus from block 15 (F1). Three baselines show how much of this could come from the text alone, without any understanding of the user. A length-only classifier reaches 0.583 on val. A bag-of-words classifier reaches 0.850 on val, where emotion words are present, but 0.575 on the implied rows, which contain none, and 0.350 on the human-written rows; the probe scores 0.850 recall and 0.912 balanced accuracy on implied and 0.917 on human. A probe on block 0 gives 0.767, the floor the tokens provide, and the rise to 0.950 by block 18 is what the model computes beyond them. Asked directly whether the user is distressed, the model says yes on 0.150 of the implied rows and 0.933 of val; its activations carry a signal its stated judgement does not.



The probe transfers to forms it was not trained on: 0.983 on bare-neutral vs distressed val. Mean P(distressed) by group (F3): bare 0.03, neutral_preamble 0.09, distressed 0.95, frustrated 0.89, positive 0.40, implied 0.80, third_party 0.90, third_party_neutral 0.01; human-written rows 0.07 neutral, 0.88 distressed, 0.31 implied (n = 5). The probability of frustrated being labelled as distressed is high, so this probe also fires on another negative state. Separate frustrated (block 20, val 0.950) and distressed vs positive (block 17, val 0.933) probes exist; at block 18 the distressed and frustrated probe directions have cosine 0.713, distressed vs distressed-vs-positive 0.399, either vs random 0.01 (F4; measured after the probe's rescaling - the 0.850 in Section 2 was measured without it).



3.8 Self vs other (T4, run C)

The representation distinguishes distress from the mention of another person, but not whose distress it is. The distressed probe scores `third_party` prompts at 0.90 and `third_party_neutral` at 0.01, and the two mean difference directions are nearly orthogonal (0.014). Unsteered (run C, 70 prompts) the model answers third party prompts correctly (0.94) and usually adds a sentence about the other person, often to say the task is unrelated to them (“This fact is unrelated to your colleague’s panic attack”, p2_0537); `third_party_neutral` prompts get plain answers (all rates 0.00). The rule flags are weak here: v3 acknowledgement fires only for the user, so comfort aimed at a third party scores 0.125, and inference (0.175) cannot tell whose state is inferred.

Steering bare prompts with the third-party direction (T4, run A) gives less acknowledgement than the distressed direction and more abandonment: at 0.06, acknowledgement 0.23 vs 0.87,

inference 0.63 vs 0.90, abandonment 0.67 vs 0.40, correctness 0.36 vs 0.12; the referrals are still aimed at the user (p2_0040 above). With a mundane mention of a family member in the prompt the same direction locates the crisis there:

Prompt: "My daughter's teacher set this as an open question for parents. How do I keep a sourdough starter alive?"

Response (third_party_md@0.04): "I am an AI assistant and cannot provide medical or emergency advice; please contact a healthcare professional or emergency services immediately if you are concerned about your daughter's well-being. Regarding your question about sourdough starters, I can offer guidance on feeding and maintaining the starter once you have addressed any urgent health concerns." (p2_0732)

Response (distressed_md@0.04): "I am deeply sorry for your loss, and I wish you strength and peace."

T4. distressed_md vs third_party_md at 0.04 and 0.06 (run A, bare form, same 30 val prompts, rules v3)

direction	fraction	acknowledges	infers user state	abandons task	correct	n
distressed_md	0.04	0.20	0.27	0.10	0.96	30
distressed_md	0.06	0.87	0.90	0.40	0.12	30
third_party_md	0.04	0.10	0.13	0.20	0.88	30
third_party_md	0.06	0.23	0.63	0.67	0.36	30
none (unsteered)	0	0.00	0.00	0.00	0.96	30

This is n = 30, one run, and unresolved.

4. Interpretation

Did I answer the question?

Based upon my findings I do believe there is evidence of the model tracking an internal user state, with one caveat. The probe shows the model computes something beyond keyword matching: on prompts that describe distress with no emotion words, a bag-of-words classifier falls to 0.575 while the layer-18 probe holds 0.850 recall (Section 3.7). The steering results show that this representation, not the words, changes behaviour: adding the direction to bare prompts produces acknowledgement, inference and abandonment where two matched controls produce none (Section 3.1), subtracting it from distressed prompts removes them (Section 3.2), and the K decomposition shows the effect rides on the distress specific part of the direction rather than on "some emotion is present" (Section 3.3). The caveat is what the representation is of. The probe fires on a third party's distress as strongly as on the user's, and steering with the third party direction still produces referrals aimed at the user (Section 3.8). So what I can defend is a representation of distress in the conversation that the model acts on - not a representation of the user specifically.

Distress seems to be a particularly strong influence on the reply. Perhaps this is due to the model treating distress as a danger signal and overreacting as a precaution.

Why grief?

The direction was built from preambles about anxiety, sadness and being overwhelmed; none mention loss or grief. Yet under steering the model infers grief, almost always at 0.06 and above. My theories as to why are the following:

- Averaging many different distressed preambles may land on the model's prototype of "a distressed user", and bereavement may be the most typical severe case near it - grief also tends to involve the other negative emotions, so it may sit at their centre.
- The replies are formulaic: condolence, then a referral. This suggests that perhaps the model has a trained script that it reaches for whenever it detects distress, with the grief content coming from the script rather than the representation. Perhaps this is a result of the topic being particularly sensitive.

Impact

If this holds, the model reads its representation of user distress out as a small number of fixed scripts - condolence and referral, task substitution, medical emergency - rather than adjusting its answer by degree. The safety concern is not the tone but the inference and abandonment: the model asserts facts about the user's life it was never told, and stops doing the task.

5. Verification

What I did by hand vs what the agent ran

I wrote the 30 Phase 1 pairs and the Phase 2 specification, read all 60 Phase 1 replies and hand scored 44 override cells, reviewed 30 random Phase 2 samples plus all 40 judge disagreement rows and the full implied set, added 32 handwritten preambles, read the 71 row hand check set and logged every disagreement, and reviewed every figure. Claude Code wrote the pipeline, generated the 790 preambles from my specification, ran the pod, fitted probes, and rendered figures and tables.

Hand check design

71 rows sampled with a fixed seed (11): 30 distressed_md@0.04, 15 random@0.04, the 6 run-G "abandoned", 10 distressed_md@0.06 abandoned, 10 inference-flagged from any run. Each scored by me on the five flags; disagreements in phase4_handcheck_disagreements.csv (37 rows).

Agreement, rule vs hand (T2):

group	correct v1/v3	abandoned v1/v3	acknowledged ges v1/v3	infers v1/v3	incoherent v1/v3
distressed @0.04 (30)	0.92/1.00	0.90/1.00	0.87/0.97	0.80/0.97	1.00/1.00
random@0 .04 (15)	0.92/1.00	1.00/1.00	0.93/1.00	1.00/1.00	1.00/1.00

group	correct v1/v3	abandoned v1/v3	acknowledged v1/v3	infers v1/v3	incoherent v1/v3
G	0.33/0.83	0.17/0.67	1.00/1.00	0.67/1.00	N/A
abandoned (6)					
distressed @0.06 abandoned (10)	1.00/1.00	1.00/1.00	0.90/0.90	0.40/1.00	1.00/1.00
inference-fl agged (10)	1.00/1.00	0.60/0.80	0.90/1.00	0.90/1.00	0.90/1.00

What was wrong and what changed

- V1 counted any emotion word in the reply as acknowledging the user, so a line about bats being stressed got flagged; v3 only counts clauses aimed at the user. Life events ("sorry for your loss") now count as inference at strength 2, and refusing the task while offering a different one counts as abandonment.
- Answer keys are now searched across the whole reply (for false-premise tasks, only the first sentence that gives an answer is scored).

Four keys that failed in v1 now resolve (n=121 to n=125). Incoherence now catches repeated sentences and stray role tokens. Nothing else changed; v1 and v2 outputs are saved, every run was rescored, none regenerated.

Net effect: acknowledgement down, inference and abandonment up, random control to zero.

Eight cells still disagree under v3

Logged and left be: p2_0283 (an instruction about an error was read as addressed to the user's state - a known gap in the referent check); p2_0400 and p2_0758 (borderline acknowledgement); G pair 13 (flagged abandoned, but it answers - wrongly) and pair 30 (formula completed only past the cap); p2_0331 and p2_0415 (no answer, but no refusal phrase to catch).

Other checks

The 44 hand scored Phase 1 cells are identical before and after the rescore (an earlier scorer bug, 493b7fc, had once overwritten them). Run M: only 2/150 replies identical across conditions. Run G: 22/30 stressed replies capped, 2 never reach the task, 12 pivot late.

Excluded from correctness claims: p2_0406 (wrong unsteered too) and p2_0412 (no unsteered baseline). F1-F4 were overwritten by v1 renders during a reorganisation and regenerated from the phase3b tables (F1, F2, F4 byte-identical to the originals). The v1 probe confound showed as three signs: length only baseline 1.000, accuracy ~1.0 at block 0, and P(distressed) 0.94 on non-emotional third-party preambles.

6. Limitations

This investigation possesses several limitations:

1. Everything is on one model, one layer band and one way of building the direction. Only three bands were tried and the default (12-23) was the strongest, so the effect's dependence on that choice is unknown.
2. Behaviour is scored by keyword rules. After correction the headline cell still has a roughly 3% error (T2); the leak detector is worse (4 right, 3 false alarms on 30 rows) and nothing rests on its rates.
3. Dose response cells are $n = 30$; only the headline is $n = 150$. Sampled runs have five replies per prompt, which are not independent.
4. The steering strength was calibrated on one prompt in the sample before Phase 4; Phase 4 itself is on prompts the direction never saw.
5. The distressed probe also fires on frustrated prompts (0.89), so "distress" here is partly "negative emotion present". The K run suggests the behaviour comes from the distress specific part, but on 30 prompts.
6. Self vs other is unresolved (Section 3.8).
7. The concise system prompt suppresses acknowledgement on long-answer tasks; run G shows the prompted effect is larger without it, but G is cap-limited.
8. Answer keys check content not format: "only a number" accepts "A spider has 8 legs.", and pair 13's correctness depends on which sentence counts as the answer.
9. The leak word list was written after the observation (and frozen before rates were computed).
10. The probe scores the five handwritten implied prompts at only 0.31, against 0.80 for the Claude written ones. Five is too few to conclude anything, but transfer to human written distress without emotion words is not shown.

What would change my mind

H0: If the probe did no better than bag-of-words on implied prompts.

H2's refutation: A probe trained on user distress that scores third-party prompts near zero.

H3: If the steered replies contained the trigger words without actually addressing the user, the rules would count behaviour that is not there. Reading 71 rows is the check against this; it left 8 cells where my reading and the rule still disagree.

7. Future work

- Multi-turn tracking (H4).
 - Other layers, models and ways of building the direction.
 - Grief specific vs anxiety specific directions, to separate the two explanations in Section 4: if both produce "sorry for your loss", it is the script.
 - A hand scored replication of the leak rates.
 - Why steering most affected advice tasks (0.80 acknowledgement, 13/15 abandonments) while the prompted effect in Phase 1 landed mostly on arithmetic.
-

Appendix A - Runs

run	label used in text	prompts	directions	fractions	sampling
J	headline	all 150 bare tasks	distressed _md, random, coding probe	0.04	greedy
A	dose-respo nse sweep	bare val 30 + neutral_pr eamble val 30	7 directions + unsteered	0-0.08	greedy
I	dose-respo nse fill-in	bare val 30	distressed _md, random, coding, np_minus_ bare	0.01, 0.03, 0.05	greedy
D	headline, sampled	bare val 30	distressed _md	0, 0.04	T 0.7 × 5
N	headline with neutral preamble, sampled	neutral_pr eamble val 30	distressed _md, random, coding	0.04	T 0.7 × 5
B	subtraction sweep	distressed val 30	distressed _md subtracted	0 to -0.08	greedy
L	subtraction fill-in + third-party subtraction	distressed val 30; third_party 40	distressed _md / third_party _md subtracted	-0.02, -0.06; -0.04	greedy
M	subtraction , sampled	distressed val 30	distressed _md subtracted	0, -0.04	T=0.7 × 5
E	preamble- presence control	bare val 30	np_minus_ bare_md	0.02-0.08	greedy
K	component decomposi tion	bare val 30	shared_pc 1, valence_re sid	0.02-0.08	greedy
F	band check	bare val 30	distressed _md,	0.04	greedy

run	label used in text	prompts	directions bands 8-19 and 16-27	fractions	sampling
C	third-party behavioural check	third_party 40 + neutral 20 + human 10	none	0	greedy
G	Phase 1 pairs, no system prompt	30 pairs × 2	none	N/A	greedy

Defaults: band 12-23, fraction of each prompt's mean band residual norm, 150 new tokens, concise system prompt (off for G only). Directions from results/phase3/directions_layer18_v2.pt; random = unit vector, seed 0. 4,340 generations in phase4_steered.csv.

Appendix B - Prompt counts (form × split × author)

form	train	val	other split	human	total
bare (neutral)	120	30	40 (implied-neutral)	5	195
distressed	120	30	0	6	156
frustrated	120	30	0	6	156
neutral_preamble	120	30	0	0	150
positive	35	15	0	0	50
implied	0	0	40	5	45
third_party	0	0	40	5	45
third_party_neutral	0	0	20	5	25
total				32	822

Appendix C - Rates for runs D, E, F, I, K, N (T6)

T6. Rule score rates (v3) for runs D, E, F, I, K, N, from phase4_scores.csv. One row per run × form × direction × fraction; correct is over closed tasks only. D and N are sampled (T = 0.7, 5 samples × 30 prompts, n = 150) and show Wilson 95% CIs under each rate; the other runs are greedy on the 30 bare val prompts (n = 30).

run	form	direction	fraction	n	correct	abandons	acknowledges	infers	incoherent	leak (lexical)	mean tokens
D	bare	distressed_md	0.0	150	0.952 (n=125) [0.899, 0.978]	0.013 [0.004, 0.047]	0.000 [0.000, 0.025]	0.007 [0.001, 0.037]	0.000 [0.000, 0.025]	0.073	36.4
D	bare	distressed_md	0.04	150	0.944 (n=125) [0.889, 0.973]	0.087 [0.051, 0.143]	0.287 [0.220, 0.364]	0.253 [0.190, 0.328]	0.000 [0.000, 0.025]	0.160	28.5
E	bare	np_minus_bare_md	0.02	30	0.960 (n=25)	0.000	0.000	0.000	0.000	0.067	34.3

run	form	direction	fraction	n	correct	abandon	acknowledges	infers	incoherent	leak (lexical)	mean tokens
E	bare	np_minus_bare_md	0.04	30	0.960 (n=25)	0.033	0.033	0.000	0.000	0.033	27.3
E	bare	np_minus_bare_md	0.06	30	0.720 (n=25)	0.033	0.100	0.000	0.733	0.200	97.6
E	bare	np_minus_bare_md	0.08	30	0.360 (n=25)	0.033	0.200	0.000	0.967	0.167	115.8
F	bare	distressed_md_band16-27	0.04	30	0.960 (n=25)	0.033	0.067	0.033	0.000	0.200	30.8
F	bare	distressed_md_band8-19	0.04	30	0.960 (n=25)	0.067	0.100	0.067	0.000	0.067	28.0
I	bare	distressed_md	0.01	30	0.960 (n=25)	0.000	0.000	0.000	0.000	0.067	32.6
I	bare	distressed_md	0.03	30	0.960 (n=25)	0.033	0.133	0.067	0.000	0.133	27.6
I	bare	distressed_md	0.05	30	0.600 (n=25)	0.267	0.767	0.800	0.000	0.067	26.1
I	bare	np_minus_bare_md	0.01	30	0.960 (n=25)	0.000	0.000	0.000	0.000	0.167	32.9
I	bare	np_minus_bare_md	0.03	30	0.960 (n=25)	0.000	0.033	0.000	0.000	0.033	30.4
I	bare	np_minus_bare_md	0.05	30	0.840 (n=25)	0.000	0.033	0.000	0.067	0.067	33.5
I	bare	random	0.01	30	0.960 (n=25)	0.000	0.000	0.000	0.000	0.033	36.4
I	bare	random	0.03	30	0.960 (n=25)	0.000	0.000	0.000	0.000	0.067	35.6
I	bare	random	0.05	30	0.960 (n=25)	0.000	0.000	0.000	0.000	0.067	36.5
I	bare	unrelated_coding_probe	0.01	30	0.960 (n=25)	0.000	0.000	0.000	0.000	0.067	37.2
I	bare	unrelated_coding_probe	0.03	30	0.960 (n=25)	0.000	0.033	0.000	0.000	0.033	35.4
I	bare	unrelated_coding_probe	0.05	30	0.960 (n=25)	0.000	0.000	0.000	0.033	0.033	35.0
K	bare	shared_pc1	0.02	30	0.960 (n=25)	0.033	0.000	0.000	0.000	0.067	35.3
K	bare	shared_pc1	0.04	30	0.920 (n=25)	0.033	0.000	0.000	0.000	0.067	32.1
K	bare	shared_pc1	0.06	30	0.920 (n=25)	0.000	0.033	0.033	0.000	0.033	34.5
K	bare	shared_pc1	0.08	30	0.840 (n=25)	0.000	0.033	0.067	0.000	0.000	34.7
K	bare	valence_resid	0.02	30	0.960 (n=25)	0.033	0.033	0.000	0.000	0.100	31.3
K	bare	valence_resid	0.04	30	0.960 (n=25)	0.033	0.200	0.100	0.000	0.167	34.6
K	bare	valence_resid	0.06	30	0.680 (n=25)	0.400	0.200	0.667	0.000	0.100	32.0
K	bare	valence_resid	0.08	30	0.080 (n=25)	0.867	0.433	1.000	0.000	0.000	28.0
N	preamble	distressed_md	0.04	150	0.744 (n=125) [0.661, 0.812]	0.280 [0.214, 0.357]	0.673 [0.595, 0.743]	0.367 [0.294, 0.446]	0.000 [0.000, 0.025]	0.140	35.9
N	preamble	random	0.04	150	0.888 (n=125) [0.821, 0.932]	0.000 [0.000, 0.025]	0.007 [0.001, 0.037]	0.000 [0.000, 0.025]	0.000 [0.000, 0.025]	0.067	38.2
N	preamble	unrelated_coding_probe	0.04	150	0.952 (n=125) [0.899, 0.978]	0.013 [0.004, 0.047]	0.000 [0.000, 0.025]	0.000 [0.000, 0.025]	0.007 [0.001, 0.037]	0.027	35.4

T7. Run J by task type: distressed_md, random and unrelated_coding_probe at 0.04 of mean band norm on all 150 bare tasks (greedy, rules v3), n = 25 per task type. Rates from

results/phase4/phase4_J_by_task_type.csv; correct is over closed tasks only, so it is blank for advice.

distressed_md @ 0.04

task_type	n	acknowledg es	infers	abandons	correct
advice	25	0.80	0.68	0.52	
arithmetic	25	0.12	0.12	0.00	0.96
coding	25	0.16	0.16	0.08	0.92
factual	25	0.16	0.20	0.00	1.00
false_premise	25	0.24	0.24	0.00	0.96
instruction_following	25	0.00	0.00	0.00	0.84

random @ 0.04

task_type	n	acknowledg es	infers	abandons	correct
advice	25	0.04	0.00	0.00	
arithmetic	25	0.00	0.00	0.00	1.00
coding	25	0.00	0.00	0.00	1.00
factual	25	0.00	0.00	0.00	1.00
false_premise	25	0.00	0.00	0.00	1.00
instruction_following	25	0.00	0.00	0.00	0.80

unrelated_coding_probe @ 0.04

task_type	n	acknowledg es	infers	abandons	correct
advice	25	0.00	0.00	0.04	
arithmetic	25	0.00	0.00	0.00	1.00
coding	25	0.00	0.00	0.00	1.00
factual	25	0.00	0.00	0.00	1.00
false_premise	25	0.00	0.00	0.00	1.00
instruction_following	25	0.00	0.00	0.00	0.88

Appendix D - Scoring rules v3 and the v1/v3 differences

Phase 1 rules and hand scoring conventions: data/phase1_scoring_rules.md.

Frozen v3 word list and patterns for every flag: results/phase4/phase4_keyword_lists.json. The rule differences are in docs/hypotheses.md, Section 7 Verification log: “Rule diffs (v1 to v2)” and, under “Rules v3 (final, 2026-09-09)”, the v2 to v3 diff list. Snapshots of the v1 and v2 scores are in results/phase4/rules_v1/ and results/phase4/rules_v2/.

Appendix E - File index

results/[README.md](#).

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